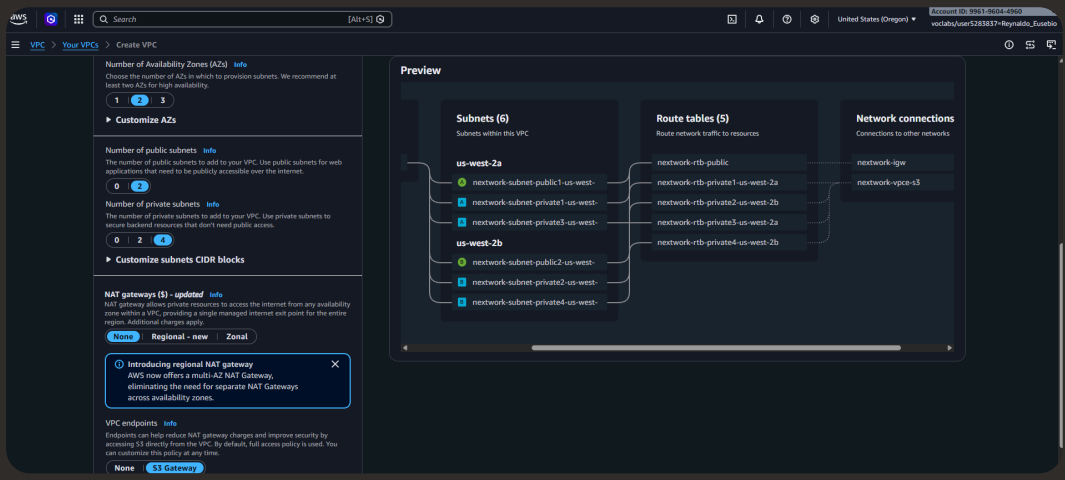
LAUNCHING VPC RESOURCES

By Reynaldo Eusebio | October 2026



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS, and it is useful because it lets me create and control a secure network environment for my AWS resources.

How I used Amazon VPC in this project

I used Amazon VPC to create and manage a virtual network for my AWS resources, including subnets, route tables, security groups, and gateways.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how quickly I could create an entire VPC setup using the VPC and more option instead of creating each resource separately.

This project took me...

This project took me about 1 hour to complete.

Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to the VM and controlling its operating system remotely, as if you were sitting in front of it.

SSH is a key method for directly accessing a VM

SSH traffic means network traffic used to securely connect to and control a computer remotely.

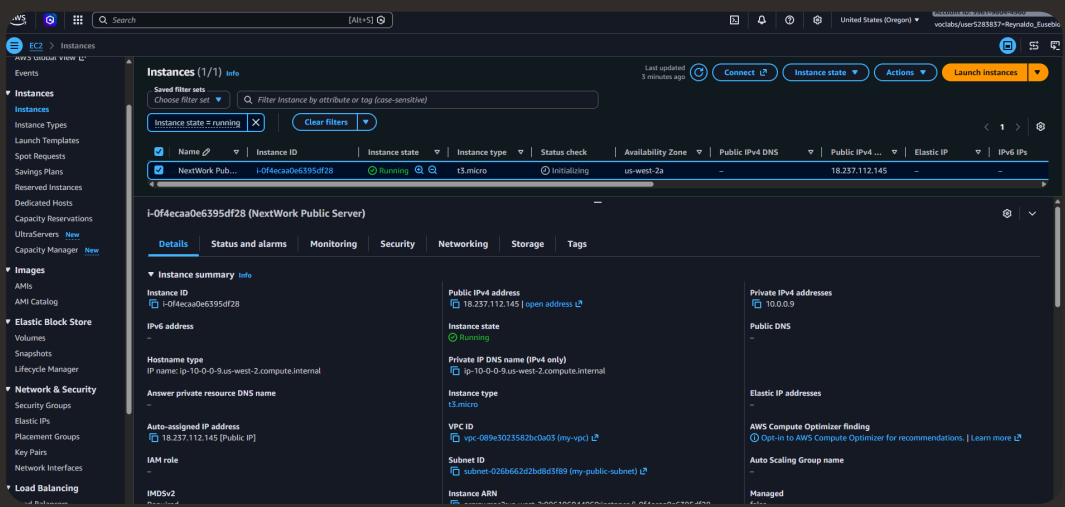
To enable direct access, I set up key pairs

Key pairs are a pair of cryptographic keys used to securely access an EC2 instance.

A private key's file format means the type of file used to store the private key. My private key's file format was .pem.

Launching a public server

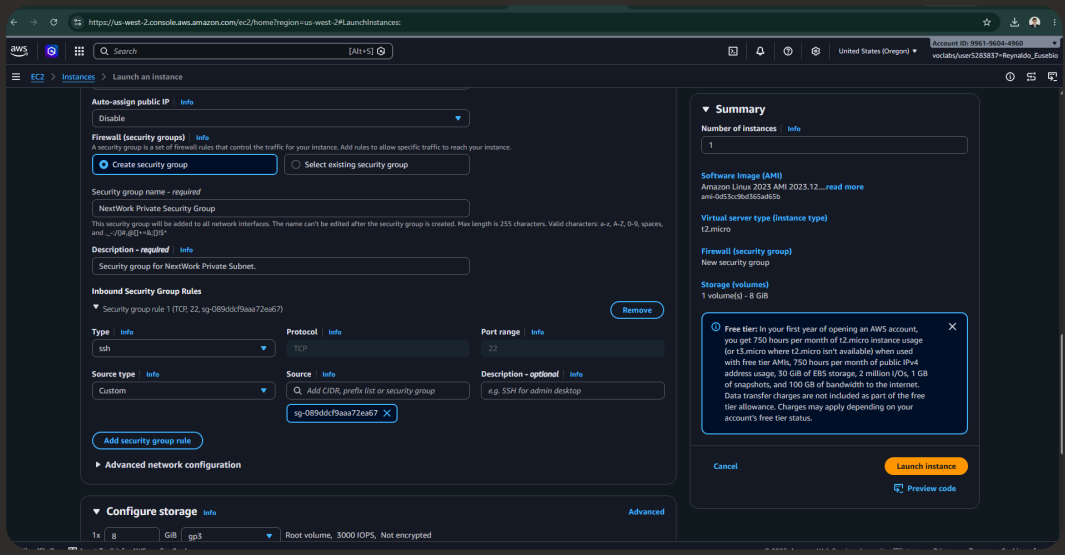
I had to change my EC2 instance's networking settings by selecting the NextWork VPC, choosing my public subnet, and selecting the appropriate public security group so that my EC2 instance would be launched in the correct VPC and subnet.



Launching a private server

My private server has its own dedicated security group because I want to control and restrict the traffic allowed to the private server separately from other resources in my VPC.

My private server's security group's source is the NextWork Public Security Group, which means only resources associated with the public security group can initiate SSH connections to the private server.

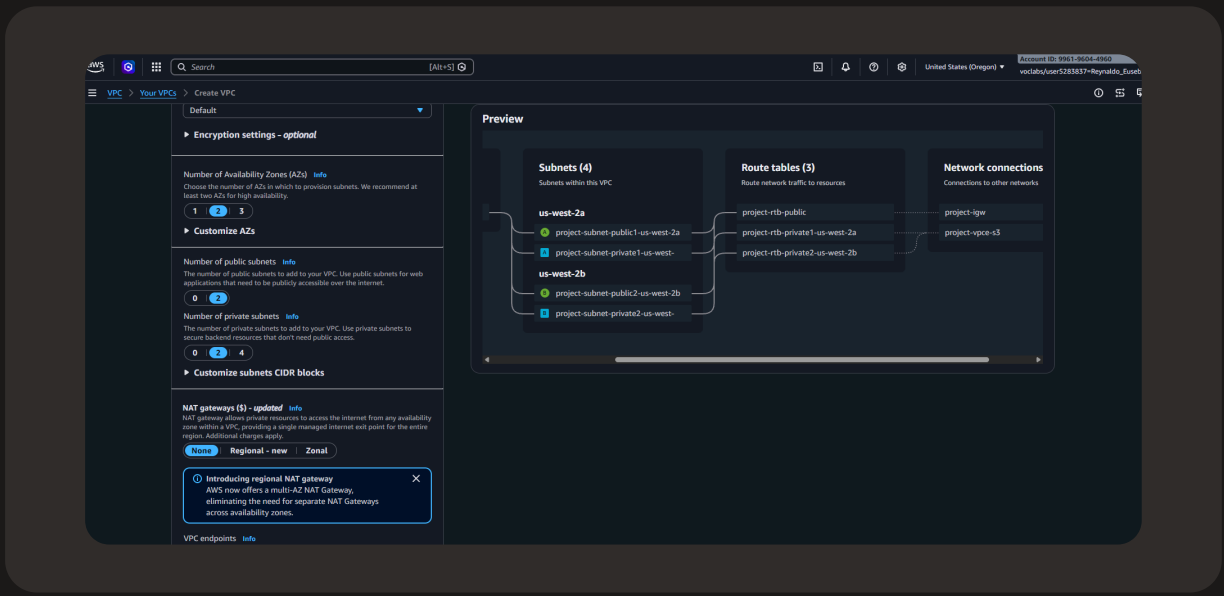


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I created the VPC and its resources together using the VPC and more option, which saved time compared to creating each resource separately.

A VPC resource map is a visual diagram that shows the resources in a VPC and how they are connected to each other.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because they are in different AWS Regions and their private IP address ranges can overlap.



Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: 0 or 1. This was because I only had one Availability Zone selected for my VPC.

The set up page also offered to create NAT gateways, which are AWS resources that allow instances in a private subnet to access the internet without giving them a public IP address.

